

Homework 10, due Nov 12 before class

1. Relativistic electron with initial Lorentz factor $\gamma_0 \gg 1$ is moving in magnetic field of value B_0 (no parallel velocity). Estimate how peak frequency of synchrotron emission evolves with time (as long as $\gamma \geq 1$). You may use approximate values, neglecting factors of $\sim$ few.
2. Circularly polarized EM wave propagating along $+z$ has electric field

$$\mathbf{E}_w = \{\sin(\omega t - kz), \cos(\omega t - kz), 0\}E_0 \quad (1)$$

Conducting plate is located at $z = 0$. (So that for $z > 0$ the total electric field is zero).

Find the reflected wave, and the total electromagnetic field in the region $z < 0$.